

What This Course Covers

The left column lists the AP Computer Science Principles framework material each unit covers. The right column lists what this course adds beyond that framework. The base course touches every AP topic at a high level; students on the optional AP track reach exam depth through a self-paced College Board-endorsed curriculum alongside class.

Touches all **35** AP CSP framework topics **32** weeks, **6** units Programming depth in **Python**

Hardware and unplugged labs throughout **Exam-free** unless a student opts in

	The AP CSP core College Board framework, all five Big Ideas	What the course adds material beyond the AP framework
WEEKS 1 TO 5 Thinking Like a Computer Scientist	BIG IDEA 2 DATA BIG IDEA 3 PROGRAMMING - Binary numbers and data compression - Variables, expressions, conditionals, loops - Procedures, libraries, random values - Finding and fixing errors	- Day one on a working 1980s Apple IIe - Computer fluency taught from zero: files, folders, and saving on a laptop, for students coming from tablets - Offline logic activities before screen work: binary cards, the human robot maze - Python installed on the student's own machine from week one
WEEKS 6 TO 10 Inside the Computer	BIG IDEA 3 PROGRAMMING - Boolean logic; lists and strings in code <i>AP treats hardware as a black box. We keep the AP programming topics moving while we open it.</i>	- Logic gates built from transistors on a breadboard - A CPU built up from NAND gates in a puzzle game - A PC disassembled in class; a bare motherboard and a Raspberry Pi examined - One keystroke traced from the key to the screen
WEEKS 11 TO 16 Programming Like a Professional	BIG IDEA 1 CREATIVE DEV BIG IDEA 3 PROGRAMMING - Data abstraction and lists - Binary search, algorithmic efficiency, undecidable problems - Simulations; testing and debugging - AP pseudocode practice begins	- Dictionaries, objects, and classes, which the AP framework does not include - Data structures acted out physically: human stacks, queues, and a floor sorting network - Writing tests for one's own code - A text-adventure build week combining the unit's material

WEEKS 17 TO
21

Operating Systems and the Internet

BIG IDEA 4 SYSTEMS & NETWORKS

- How the internet works: addressing, routing, protocols
- Fault tolerance and redundancy

- The terminal: the macOS Unix shell, and Linux on the Windows machines
- A physical network built in class with a switch and router
- Live packet captures in Wireshark, including what HTTPS encrypts
- SSH into the classroom's own server
- Mid-year milestone: tracing a button press end to end, orally

WEEKS 22 TO
26

Building Modern Software

BIG IDEA 1 CREATIVE DEV

BIG IDEA 2 DATA

- Collaboration and program design
- Extracting information from data; using programs with data
- Libraries in real use

- VS Code, and Git with merge conflicts resolved on the class server
- A web page written from scratch: HTML, CSS, JavaScript
- A Python web app calling a live weather API
- A public dataset analyzed; the app converted to an installable PWA

WEEKS 27 TO
32

The Future of Computing

BIG IDEA 5 IMPACT

BIG IDEA 4 SYSTEMS & NETWORKS

- Safe computing and cybersecurity concepts
- All six Impact of Computing topics: benefit and harm, bias, the digital divide, crowdsourcing, law and ethics
- Parallel and distributed computing
- The AP Create Performance Task, with its nine supervised hours, for opted-in students

- AI from a hand-built perceptron to how large language models work
- Responsible AI use taught explicitly, then permitted; until that unit, students write every line themselves
- Cryptography through physical activities: cipher wheels, paint-mixing key exchange
- Cloud, containers, and blockchain covered in a single overview lesson
- A final project of the student's own design, demonstrated to the class

Students work on a real local computer, not a browser sandbox: they install Python, manage files, use the terminal, capture network traffic, and version their code with Git on the same laptop all year. The computer itself is part of the curriculum.

About the AP track: the AP layer is opt-in per student and exam-free by default. Students who opt in prepare through a free, College Board-endorsed online curriculum alongside class and may sit the AP CSP exam in May; everyone else takes the same course and earns a full Computer Science I credit with no exam. The base course touches every AP topic at a high level; touching a topic is not the same as teaching it to exam depth, and exam depth is what the AP track's self-paced curriculum provides. Coverage is documented topic by topic in the course's coverage map.

Details of outside services, AP requirements, fees, and dates change; verify current specifics with the instructor.